

GRAPHING QUADRATICS — Cheatsheet

Algebra 1 · Topics 7–9 + 14 (Standard Form) · Esme · pre-approved by Mr. Mistry

★ THE 4 FORMS — WHAT EACH SHOWS

Form	Equation	Free info
Basic	$y = ax^2$	vertex (0,0)
Vertex	$y = a(x-h)^2 + k$	vertex (h, k)
Intercept	$y = a(x-p)(x-q)$	roots p, q
Standard	$y = ax^2 + bx + c$	y-int (0, c)

★ ax^2 HEIGHTS PATTERN

From vertex, the y-jumps left/right are:

Steps	± 1	± 2	± 3
y-jump	a·1	a·4	a·9
a=1	1	4	9
a=2	2	8	18
a=½	0.5	2	4.5

Memorize 1, 4, 9. Multiply by a for any parabola.

VERTEX FORM — $y = a(x - h)^2 + k$

Vertex form gives you the vertex for free.

- Read vertex (h, k). **SIGN OF h FLIPS!**
- AOS: $x = h$.
- Direction: $a > 0$ up, $a < 0$ down.
- Plot vertex, use heights pattern ($\times a$) for more points.
- Label vertex + AOS.

$y = 2(x-3)^2 + 1 \rightarrow$ vertex (3, 1),
AOS $x=3$, up
 $(x+5)^2 \rightarrow h = -5$ (flip!)

Hidden inside flips. Kept outside stays.

INTERCEPT FORM — $y = a(x - p)(x - q)$

Gives you the x-intercepts (roots) for free.

- Read roots p and q. **SIGN FLIPS** ($x-5 \rightarrow$ root +5).
- Plot (p, 0) and (q, 0).
- AOS = $(p + q) / 2$.
- Plug AOS x into eq $\rightarrow$ vertex y.
- Plot vertex. Smooth U.

$y = (x-1)(x-5)$
roots 1, 5 $\rightarrow$ AOS $x=3 \rightarrow$ vertex (3, -4)

★ STANDARD FORM — $y = ax^2 + bx + c$ (NEW)

Vertex is HIDDEN — compute it. Most common on tests.

- AOS:** $x = -b / (2a)$. **Watch sign of b.**
- Plug AOS x into eq $\rightarrow$ vertex y.
- y-int is (0, c). Mirror it across AOS $\rightarrow$ free 2nd point.
- Use heights pattern ($\times a$) for more points.
- Smooth U. Label vertex + AOS.
- (Optional) x-ints: factor / QF / complete square.

$y = x^2 - 6x + 5$
AOS: $x = 6/2 = 3$
Vertex y: $9 - 18 + 5 = -4 \rightarrow (3, -4)$
y-int (0, 5), mirror (6, 5)
Heights (a=1): (2, -3), (4, -3);
(1, 0), (5, 0)

Mnemonic: AOS, VERTEX, c, HEIGHTS.

BASIC — $y = ax^2$

- Vertex (0, 0). AOS $x = 0$.
- Direction: $a > 0$ smile, $a < 0$ frown.
- Width: $|a| > 1$ narrower, $|a| < 1$ wider.
- Table $x = -2, -1, 0, 1, 2$ OR heights pattern.

$y = 2x^2$: heights 2, 8 $\rightarrow (1, 2)$,
(2, 8) and mirrors

The Role of "a"

a	Effect
$a > 0$	opens UP (smile) — has minimum
$a < 0$	opens DOWN (frown) — has maximum
$ a > 1$	NARROWER than $y = x^2$
$ a < 1$	WIDER (flattened)
$ a = 1$	standard width

Bigger a = SKINNIER (counter-intuitive!)

Convert Between Forms

From $\rightarrow$ To	How
Vertex $\rightarrow$ Standard	Expand $(x-h)^2$
Intercept $\rightarrow$ Standard	FOIL it out
Standard $\rightarrow$ Vertex	Complete the square (Topic 12)
Standard $\rightarrow$ Intercept	Factor (Topics 4/5)

Vocab

Parabola	U-shaped curve
Vertex	Tip of U (min or max)
AOS	Vertical line through vertex
Domain	All real numbers
Range	$y \geq k$ (up) or $y \leq k$ (down)
x-int / root / zero	Same thing: where $y = 0$
y-int	(0, c) in standard form

★ FULL-CREDIT GRAPH CHECKLIST

- ☐ Vertex plotted AND labeled (h, k)
- ☐ AOS drawn (dashed line) AND labeled ($x = h$)
- ☐ At least 5 points plotted
- ☐ Smooth U (not a V!)
- ☐ Direction correct (smile vs frown)
- ☐ x-intercepts labeled (if applicable)
- ☐ y-intercept labeled (if visible)

⚠ TOP TRAPS

- Sign of h:** $(x-3)^2 \rightarrow h=+3$; $(x+3)^2 \rightarrow h=-3$.
- Sign of b in AOS:** $b=-6 \rightarrow x = -(-6)/2 = +3$.
- Drew a V:** parabolas have rounded vertices.
- Forgot to label AOS:** rubric wants it.
- Bigger a = NARROWER** (counter-intuitive).

Worked Ex — Standard Form

Graph: $y = x^2 - 4x + 3$

- a=1, b=-4, c=3
- AOS: $x = 4/2 = 2$
- Vertex y: $4 - 8 + 3 = -1 \rightarrow (2, -1)$
- y-int (0, 3); mirror (4, 3)
- Heights (a=1): (1, 0), (3, 0) [x-intercepts!]
- Smooth U up, label.